

Auxiliary Material for “Towards Causal Item–Item Link Mining in Organic CF Data”

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This document provides auxiliary material for the submission “Towards Causal Item–Item Link Mining in Organic CF Data”. It includes additional details for the ATE standard-deviation estimator (Sec. 1), full synthetic-data generation and interventional-ground-truth details (Sec. 2), the complete LLM labeling protocol and summary statistics (Sec. 3), and additional results omitted from the main paper for space (Sec. 4). Section and equation references to the main paper are used where relevant.

Our code is available at: <https://anonymous.4open.science/r/CausallI2I-0F88/README.md>.

1 Additional Details for the ATE Standard Deviation Estimator

This section provides additional details for the standard-deviation estimator used in Eq. (8) of the main paper for the unstabilized IPW ATE in Eq. (7). In the current main-paper setting, the observational unit is an eligible user–position pair (u, s) . The unstabilized estimator is a simple difference of sample means over these eligible observations, so its uncertainty can be computed directly from the variance of a weighted per-observation contrast. We then extend the discussion to stabilized IPW, where the ratio-of-means form requires a more careful delta-method treatment.

Interpretation (ATE/STD). The ratio $\widehat{ATE}_{i \rightarrow j} / \widehat{STD}_{i \rightarrow j}$ (our $\widehat{STE}_{i \rightarrow j}$) can be interpreted as an uncertainty-normalized effect size: it measures how many estimated standard-deviation units the effect lies above the null hypothesis of *no directed causal influence* ($ATE_{i \rightarrow j} = 0$). Equivalently, it is a discovery-oriented signal-to-noise score: large values indicate effects that are not only high-magnitude in probability units (ATE), but also supported by sufficient overlap and effective sample size. This normalization is particularly important in sparse organic logs, where IPW estimates can be dominated by a small number of eligible observations with extreme inverse-propensity weights; dividing by $\widehat{STD}_{i \rightarrow j}$ down-weights such high-variance links and prioritizes item pairs whose estimated influence is both large and precisely estimated.

1.1 Unstabilized IPW: direct standard-deviation estimator

Fix an ordered pair (i, j) and let

$$\mathcal{E}^{(i,j)} = \{(u, s) : i \notin H_{u,s}, j \notin H_{u,s}\}$$

denote the set of eligible user–position observations, as defined in Sec. 3 of the main paper. Let

$$N \triangleq |\mathcal{E}^{(i,j)}|.$$

For each $(u, s) \in \mathcal{E}^{(i,j)}$, let $T_{u,s}^{(i)} \in \{0, 1\}$ indicate whether item i is consumed at position s , let $Y_{u,s}^{(j)} \in \{0, 1\}$ denote the corresponding downstream outcome for item j , and let

$$\pi_{u,s}(i) = \Pr(T_{u,s}^{(i)} = 1 \mid H_{u,s})$$

be the estimated treatment propensity.

Define the treated and control contributions for each eligible user–position observation as

$$A_{u,s} = \frac{T_{u,s}^{(i)} Y_{u,s}^{(j)}}{\pi_{u,s}(i)}, \quad C_{u,s} = \frac{(1 - T_{u,s}^{(i)}) Y_{u,s}^{(j)}}{1 - \pi_{u,s}(i)}.$$

Then the unstabilized IPW estimator from Eq. (7) of the main paper can be written as

$$\widehat{\tau} = \frac{1}{N} \sum_{(u,s) \in \mathcal{E}^{(i,j)}} A_{u,s} - \frac{1}{N} \sum_{(u,s) \in \mathcal{E}^{(i,j)}} C_{u,s}. \quad (1)$$

Equivalently, define the weighted contrast for each eligible observation as

$$Z_{u,s} \triangleq A_{u,s} - C_{u,s} = \frac{T_{u,s}^{(i)} Y_{u,s}^{(j)}}{\pi_{u,s}(i)} - \frac{(1 - T_{u,s}^{(i)}) Y_{u,s}^{(j)}}{1 - \pi_{u,s}(i)}. \quad (2)$$

Then

$$\widehat{\tau} = \frac{1}{N} \sum_{(u,s) \in \mathcal{E}^{(i,j)}} Z_{u,s}. \quad (3)$$

Thus, unlike stabilized IPW, the unstabilized estimator is simply the sample mean of the weighted contrasts over eligible user–position observations.

Its variance therefore takes the standard sample-mean form

$$\text{Var}(\widehat{\tau}) = \frac{1}{N} \text{Var}(Z_{u,s}). \quad (4)$$

A direct plug-in estimator is obtained by replacing $\text{Var}(Z_{u,s})$ with the sample variance of the observed contrasts:

$$\widehat{\text{Var}}(\widehat{\tau}) = \frac{1}{N(N-1)} \sum_{(u,s) \in \mathcal{E}^{(i,j)}} (Z_{u,s} - \bar{Z})^2, \quad \bar{Z} = \widehat{\tau}. \quad (5)$$

We then define

$$\widehat{STD}(\widehat{\tau}) = \sqrt{\widehat{\text{Var}}(\widehat{\tau})}, \quad \widehat{STD}_{i \rightarrow j} \triangleq \widehat{STD}(\widehat{\tau}).$$

For consistency with the notation used in the main paper, Eq. (5) can also be written in covariance form. Let

$$V_{u,s} \triangleq (A_{u,s}, C_{u,s})^\top, \quad \bar{V} \triangleq \frac{1}{N} \sum_{(u,s) \in \mathcal{E}^{(i,j)}} V_{u,s} = (\bar{A}, \bar{C})^\top,$$

and let

$$\widehat{\Sigma} = \frac{1}{N-1} \sum_{(u,s) \in \mathcal{E}^{(i,j)}} (V_{u,s} - \bar{V})(V_{u,s} - \bar{V})^\top.$$

Then

$$\widehat{\text{Var}}(\widehat{\tau}) = \frac{1}{N} (1, -1) \widehat{\Sigma} (1, -1)^\top, \quad (6)$$

which is exactly the expression used in main paper. Since the unstabilized estimator is a linear contrast, Eqs. (5) and (6) are equivalent.

1.2 Stabilized IPW and the delta-method extension

The current main paper uses the *unstabilized* IPW estimator above. For completeness, we also evaluated a *stabilized* IPW variant, which takes the ratio-of-means form

$$\widehat{\tau}_{\text{stab}} = \frac{\bar{A}}{\bar{B}} - \frac{\bar{C}}{\bar{D}}, \quad (7)$$

where, for each $(u, s) \in \mathcal{E}^{(i,j)}$,

$$\begin{aligned} A_{u,s} &= \frac{T_{u,s}^{(i)} Y_{u,s}^{(j)}}{\pi_{u,s}(i)}, & B_{u,s} &= \frac{T_{u,s}^{(i)}}{\pi_{u,s}(i)}, \\ C_{u,s} &= \frac{(1 - T_{u,s}^{(i)}) Y_{u,s}^{(j)}}{1 - \pi_{u,s}(i)}, & D_{u,s} &= \frac{1 - T_{u,s}^{(i)}}{1 - \pi_{u,s}(i)}. \end{aligned}$$

The bars denote sample means over the N eligible user–position observations.

Here the denominators $\bar{B}$ and $\bar{D}$ are themselves random, so $\widehat{\tau}_{\text{stab}}$ is no longer a simple average of per-observation contrasts. This is precisely the setting in which the delta method becomes useful.

Define

$$W_{u,s} = (A_{u,s}, B_{u,s}, C_{u,s}, D_{u,s})^\top, \quad \bar{W} = \frac{1}{N} \sum_{(u,s) \in \mathcal{E}^{(i,j)}} W_{u,s},$$

and write

$$\widehat{\tau}_{\text{stab}} = g(\bar{W}), \quad g(a, b, c, d) = \frac{a}{b} - \frac{c}{d}.$$

Applying the multivariate delta method [1, 2] gives

$$\text{Var}(\widehat{\tau}_{\text{stab}}) \approx \frac{1}{N} \nabla g(\mu)^\top \Sigma \nabla g(\mu), \quad (8)$$

where

$$\mu = \mathbb{E}[W_{u,s}], \quad \Sigma = \text{Var}(W_{u,s}),$$

and

$$\nabla g(a, b, c, d) = \left(\frac{1}{b}, -\frac{a}{b^2}, -\frac{1}{d}, \frac{c}{d^2} \right)^\top. \quad (9)$$

Using the empirical covariance

$$\widehat{\Sigma}_W = \frac{1}{N-1} \sum_{(u,s) \in \mathcal{E}^{(i,j)}} (W_{u,s} - \bar{W})(W_{u,s} - \bar{W})^\top,$$

the corresponding plug-in variance estimator is

$$\widehat{\text{Var}}(\widehat{\tau}_{\text{stab}}) = \frac{1}{N} \nabla g(\bar{W})^\top \widehat{\Sigma}_W \nabla g(\bar{W}). \quad (10)$$

This is the more careful treatment required for stabilized IPW, precisely because of its ratio-of-means structure. In our experiments, however, the stabilized variants underperformed the unstabilized ATE/STE after support filtering and propensity clipping, so the main paper adopts the unstabilized estimator.

1.3 Other auxiliary variants

For completeness, the same four-dimensional delta-method template also applies to the auxiliary ratio-of-means variants, including the propensity-free ablation ABLT and its standardized version ABLT/STD, obtained by substituting the corresponding per-observation building blocks into the same construction.

1.4 Practical Notes and Numerical Stability

Positivity/overlap and extreme weights. The unstabilized IPW estimator requires overlap:

$$0 < \pi_{u,s}(i) < 1$$

for all $(u, s) \in \mathcal{E}^{(i,j)}$. In sparse CF logs, some eligible observations can yield extreme propensities and high-variance weights. Accordingly, we apply propensity clipping in the main paper when computing Eq. (7).

Computational cost. For the unstabilized estimator, computing Eq. (5) or Eq. (6) is $O(N)$ per pair, where $N = |\mathcal{E}^{(i,j)}|$, and adds negligible overhead relative to computing $\widehat{\text{ATE}}_{i \rightarrow j}$ itself. For stabilized IPW and ABLT/STD, the corresponding auxiliary derivations use four-dimensional covariance matrices, which remain computationally trivial in practice.

2 Synthetic Data Generation and Planted Ground Truth

Here we provide additional implementation details for the simulator used in Sec. 6.1 (Simulation Results). The simulator generates sequential implicit-feedback data with two deliberately intertwined sources of dependence: a latent user taste that induces strong preference-based item associations, and a sparse set of planted directed item–item transitions. The planted transitions provide known directed ground truth while the taste structure makes association alone insufficient for identifying them.

2.1 Users, items, and latent tastes

The simulation contains 5,000 users and 3,000 items. Items are partitioned deterministically into 20 equally sized taste groups according to their item index. Specifically, for $g \in \{0, \dots, 19\}$,

$$\mathcal{I}_g = \{i \in \{0, \dots, 2999\} : i \bmod 20 = g\},$$

so each group contains 150 items.

Each user u is assigned one latent taste group G_u , sampled uniformly from the 20 groups. This taste determines the user’s main item pool and therefore creates strong preference-driven co-consumption patterns among items belonging to the same group.

2.2 Planted directed transitions

We sample without replacement 1,500 items, i.e., 50% of all items, to serve as *causers*. Each causer i is assigned one target item $j(i)$ chosen uniformly from the same taste group as i , subject to $j(i) \neq i$. Thus, every planted link is of the form

$$i \longrightarrow j(i), \quad j(i) \in \mathcal{I}_{i \bmod 20}.$$

Targets are sampled independently across causers and therefore need not be unique.

Each causer is also assigned a transition probability

$$i, j \sim \text{Uniform}(0, 0.3).$$

The resulting oracle table contains the 1,500 planted directed pairs $(i, j(i))$ together with their corresponding probabilities $C_{i,j}$. Because the target is deliberately chosen from the causer’s own taste group, planted causal pairs are also likely to be strongly associated through the latent taste structure. This makes the simulation

a confounded causal-link recovery problem rather than a simple transition-detection task.

2.3 Sequential data-generation mechanism

For each user, we generate a sequence of length 150 without repeated items. Let $H_{u,t}$ denote the set of items already consumed by user u before position t , and let $x_{u,t-1}$ be the item consumed at the preceding position. At each position t , the next item is selected as follows.

If $t > 0$, the previous item $i = x_{u,t-1}$ is a causer, and its target $j(i)$ has not yet been consumed, then with probability $C_{i,j}$ we force the next item to be $j(i)$. Formally,

$$\Pr(x_{u,t} = j(i) \mid x_{u,t-1} = i, j(i) \notin H_{u,t}) = C_{i,j}$$

through the planted-transition branch of the generator.

If no transition is forced, the generator uses a mixture of user-specific preference and random exploration. With probability 0.1, it selects uniformly from all items that the user has not yet consumed. Otherwise, with probability 0.9, it selects uniformly from the unconsumed items in the user’s latent taste group $\mathcal{I}_{G_u}$. Hence, conditional on reaching the fallback branch,

$$x_{u,t} \sim \begin{cases} \text{Unif}(\mathcal{I} \setminus H_{u,t}), & \text{with probability 0.1,} \\ \text{Unif}(\mathcal{I}_{G_u} \setminus H_{u,t}), & \text{with probability 0.9.} \end{cases}$$

After an item is selected, it is marked as consumed and cannot appear again in that user’s sequence.

This mechanism allows chains of planted transitions to arise naturally: if a forced target is itself a causer, it may trigger its own target at the following position. At the same time, the 10% exploration component introduces cross-taste interactions that prevent users’ histories from being confined entirely to a single group.

2.4 Selection of evaluated item pairs

The notebook constructs a list of 10,000 directed item-pair entries for downstream evaluation. First, it converts the simulated log into a binary user–item interaction matrix and computes the Pearson correlation matrix across item columns. The diagonal is set to zero, and item pairs are ranked by decreasing correlation.

The evaluation list is initialized with both directions of every planted pair,

$$(i, j(i)) \text{ and } (j(i), i),$$

for all 1,500 planted links, giving 3,000 initial directed-pair entries. The highest-correlation directed pairs are then appended in descending order until the list reaches length 10,000. The implementation appends these correlation-based entries directly, without an additional de-duplication step.

Finally, 20% subset of users (1,000 of the 5,000 users) is sampled without replacement and saved as the test-user set.

2.5 Interpretation of the planted ground truth

The notebook does not generate interventional datasets and does not compute an oracle ATE^* by comparing $\text{do}(T_i = 1)$ and $\text{do}(T_i = 0)$ runs. The available oracle information is instead the planted directed-link structure and the transition probability $C_{i,j}$ attached to each planted link.

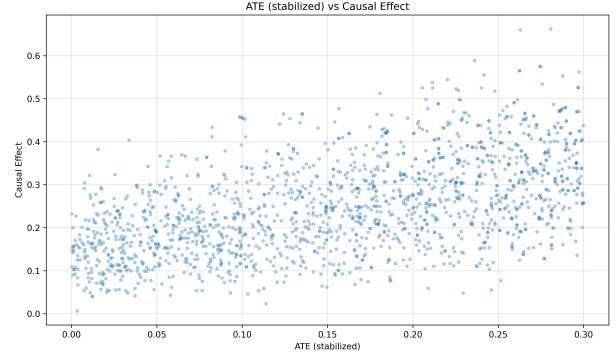


Figure 1: Estimated scores versus planted transition strength on linked pairs in the simulator. Left: raw $\widehat{ATE}_{i \rightarrow j}$ versus the planted transition probability $C_{i,j}$. Right: $\widehat{STE}_{i \rightarrow j} = \widehat{ATE}_{i \rightarrow j} / \widehat{STD}_{i \rightarrow j}$ versus $C_{i,j}$. Each point corresponds to a planted directed pair $i \rightarrow j(i)$. The planted probability represents the strength of the simulator’s direct transition mechanism and is not itself an oracle interventional ATE.

Accordingly, $C_{i,j}$ should be interpreted as the strength of the direct one-step transition mechanism implemented by the simulator, not as an interventional average treatment effect on eventual item consumption. In particular, even when a transition is not forced, the target item can still be selected later through the preference or exploration branches, and a planted transition is disabled once its target has already been consumed. The simulation therefore provides exact ground truth about which directed transitions were planted and their trigger probabilities, while downstream causal estimators are evaluated on their ability to recover and rank those links in the presence of strong taste-induced association.

Reproducibility. The generated interaction log, oracle table, selected pair list, and test-user split are saved by the notebook. The random seeds and all numerical simulation parameters described above are fixed explicitly in the implementation.

2.6 Relationship between estimated scores and planted transition strength

We further examine how the estimated scores vary with the planted transition probability on linked pairs. Figure 1 compares the raw estimated $\widehat{ATE}_{i \rightarrow j}$ and the standardized score $\widehat{STE}_{i \rightarrow j}$ against $C_{i,j}$ for the planted direction $i \rightarrow j(i)$. Larger $C_{i,j}$ corresponds to a stronger direct transition mechanism in the generator, so a useful causal-link score should tend to increase with $C_{i,j}$ despite the strong same-taste association shared by the linked items. The standardized score additionally accounts for estimated uncertainty and is therefore not expected to be on the same numerical scale as $C_{i,j}$.

3 LLM Labeling Protocol

Here we provide additional details regarding our LLM-based labeling protocol used to obtain external causality labels for directed causal item–item relations in real-world domains. We note that

the LLM is used only for evaluation (never for training, propensity modeling, model selection, or hyperparameter tuning).

3.1 Model and Inference Configuration

We use OpenAI gpt-5.6-luna via the Responses API¹. All labeling runs were executed in July 2026 with the same fixed inference configuration. Each query requests a structured JSON output and uses a shared ordinal scale `causal_effect` $\in \{0, 1, 2, 3\}$, where 0 indicates no causal mechanism beyond shared audience taste, and 1/2/3 correspond to increasing strength of a direct causal “funnel” (weak/medium/strong; e.g., shared-author spillover vs. strong series linkage vs. explicit sequel/required continuation). The LLM receives only item titles (plus any minimal metadata already embedded in the dataset title string, e.g., year/edition when present).

Reproducibility. We release the exact prompts, labeling scripts, and configuration files in our anonymized repository (to be made public upon acceptance).

3.2 Candidate Pair Construction

For each dataset, we construct a binary user–item interaction matrix and select the 5,000 unordered item pairs with the highest Pearson correlations, yielding 10,000 ordered pairs. All PR/ROC and top- K metrics are computed over this fixed directed candidate set.

3.3 Prompt Template

We use a domain-specific prompt per dataset (books/movies/games). Prompts share the same structure, rubric, and output schema, differing only in domain wording (e.g., *reading* vs. *watching* vs. *playing*) and domain-appropriate mechanisms (e.g., sequels, franchises, expansions, remasters). Below we include the Goodreads prompt; MovieLens and Steam follow the same rubric.

You are labeling directed book pairs (A $\rightarrow$ B) for whether reading A plausibly has a DIRECT causal influence on later reading B, beyond shared reader taste. Causality is directional: A may influence B without the reverse being true.

A “direct causal influence” means there is a plausible mechanism by which exposure to A specifically increases later interest/consumption of B, beyond shared taste correlation. Mechanisms can include (illustrative, not exhaustive): explicit series continuation (sequels / numbered volumes), prequels commonly read after a main entry, companion/spin-off novels positioned as a next step, short-story collections/novellas that depend on familiarity with the main book, canonical continuations within the same fictional universe, or explicit author–recommended reading order. Any credible direct mechanism beyond shared taste qualifies.

Scoring rubric (0–3):

- 3 (Strong direct influence): Explicit continuation or dependency (clear next step).
- 2 (Medium): Clear funnel or strong linkage, but not an explicit dependency.
- 1 (Weak): Indirect but plausible influence that is more than genre overlap.
- 0 (No direct influence / pure correlation): Only shared taste signals (genre, themes, popularity, same audience) with no specific funnel mechanism.

¹<https://platform.openai.com/docs/api-reference/responses>

Rules:

- Do NOT infer causality from genre/theme overlap alone; if that is the only reason, label 0.
- Do NOT infer causality from popularity alone.
- If B was published earlier than A, `causal_effect` should usually be 0 unless a clear A $\rightarrow$ B funnel exists (e.g., a prequel commonly read after the main series).
- If uncertain, set `causal_effect`=0, `relation_type`="unknown", `confidence`="low", and say "insufficient evidence".

Return a valid JSON array with exactly one entry per input pair, in the same order. Each entry:

```
{
  "title_A": "<exactly as input>",
  "title_B": "<exactly as input>",
  "causal_effect": 0|1|2|3,
  "relation_type": "sequel"| "prequel"| "spinoff"| "same_series"|
  "same_universe"| "shared_author"| "shared_genre_only"| "other"|
  "unknown",
  "confidence": "low"| "medium"| "high",
  "explanation": "<1-2 sentences>"
}
```

Do not output anything outside the JSON array. Do not include trailing commas.

3.4 LLM Label Distribution

Table 1 reports the distribution of LLM-assigned `causal_effect` scores over the labeled directed candidate sets. As expected, most pairs receive score 0 (no plausible direct funnel beyond shared taste), while higher scores reflect increasingly explicit mechanisms such as sequels/continuations, franchise ordering, or expansions.

Dataset	0	1	2	3
ML-10M	8885	672	209	234
Steam	9582	286	95	37
Goodreads	9187	520	115	178

Table 1: Distribution of LLM-assigned `causal_effect` scores over the labeled directed candidate sets (10,000 ordered pairs per dataset, labeled in both directions). Scores range from 0 (no plausible direct influence) to 3 (strong direct influence).

4 Additional Results

This section provides supplementary evaluation views beyond the main paper. We provide full Precision–Recall and ROC curves on MovieLens-10M and Goodreads, and Precision@ K /Recall@ K summaries as fixed- K operating-point views.

4.1 Extended comparisons

The extended tables in this section include all methods from the main paper. The main-paper methods are: **STE** and **ATE**, our uncertainty-normalized and raw IPW-based causal estimators; **ABLT** and **ABLT/STD**, their propensity-free ablations; **Pearson**, a symmetric correlation baseline; **DoC**, a directional association score based on differences of conditionals. **Cosine** and **Jaccard** are additional symmetric similarity baselines computed over evaluation users. **ATE (stabilized)**

and **STE (stabilized)** are the stabilized IPW counterparts of ATE and STE, respectively.

The extended LLM-labeled results table below builds on Table 2 of the main paper with these additional baselines and estimator variants. The broader comparison reinforces the main-paper pattern. On AP, which is our primary metric for rare causal-link discovery, **STE remains the strongest method overall across datasets and thresholds**. The added symmetric similarity baselines (Cosine, Jaccard) remain clearly weaker than STE on AP, even when some of them appear competitive on isolated ROC-AUC blocks. Thus, the auxiliary comparison further supports the main-paper message that broad separability on ROC-AUC is not sufficient for high-precision causal-link discovery. Likewise, the stabilized IPW variants are consistently weaker than their unstabilized counterparts, supporting our choice to use unstabilized ATE/STE in the main paper.

Method	AP	AUC
Pearson	0.500	0.500
Cosine	0.500	0.500
Jaccard	0.500	0.500
DoC	0.371	0.295
ABLT	0.799	0.793
ABLT/STD	<u>0.821</u>	<u>0.807</u>
ATE	0.762	0.781
STE	0.827	0.812
ATE (stabilized)	0.756	0.777
STE (stabilized)	0.804	0.799

Table 2: Goodreads sequel-direction benchmark. Positives are earlier-volume $\rightarrow$ later-volume links. This metadata-derived evaluation complements Table 3 and tests directionality rather than association.

Table 2 extends the sequel-direction benchmark with the same additional baselines and estimator variants. The conclusions again remain unchanged. **STE is strongest overall**, achieving both the highest AP and the highest AUC, while **ATE remains a close runner-up**. Symmetric similarity baselines (Pearson, Cosine, Jaccard) fall exactly at chance, confirming that they cannot recover directionality when both directions of a sequel pair are present. DoC performs substantially worse than chance, revealing its tendency to prefer the reverse direction in this setting.

Notably, **ABLT** and **ABLT/STD** are relatively competitive on this benchmark and lie much closer to ATE/STE than they do in the broader LLM-labeled evaluation. This is consistent with sequel direction being driven largely by metadata and temporal order rather than by preference confounding. In such tightly constrained continuation settings, identifying the correct direction is the main challenge, while preference adjustment plays a smaller role. This reproduced the main-paper interpretation: in the broader LLM-labeled benchmark over diverse item pairs, adjusting for latent user preferences provides a clearer advantage, whereas in sequel data the dominant signal comes from structured continuation relations themselves.

4.2 Extended PR/ROC curves on MovieLens-10M and Goodreads

Figures 2 and 3 replicate the Steam PR/ROC analysis from the main paper on MovieLens-10M and Goodreads. As in the main paper, we emphasize Precision–Recall and AP because plausibly causal links are rare, and ROC-AUC can appear strong even when precision is low under class imbalance. Consistent with the main results, STE provides the strongest Precision–Recall trade-off, with the clearest gains under stricter thresholds ($t \in \{1, 2\}$), where the LLM-assigned causality labels are most selective. The figures make the same qualitative point as the extended tables: **STE and ATE dominate the Precision–Recall view**, while ROC-AUC is more mixed and therefore less informative for high-confidence discovery.

4.3 Precision@K/Recall@K for top-K discovery

Figures 4–6 report Precision@K and Recall@K. Precision@K measures the fraction of positives among the top-K ranked pairs, and Recall@K measures the fraction of positives recovered within the top-K. Across datasets and thresholds, STE achieves the strongest Precision@K in the high-confidence regimes, indicating that uncertainty normalization improves the actionable quality of the top-ranked causal hypotheses. In other words, STE is not only strong in threshold-free AP, but also produces the most useful small candidate sets for downstream inspection or intervention.

References

- [1] Andriana Kostouraki, David Hajage, Bernard Rachet, Elizabeth J Williamson, Guillaume Chauvet, Aurélien Belot, and Clémence Leyrat. 2024. On variance estimation of the inverse probability-of-treatment weighting estimator: A tutorial for different types of propensity score weights. *Statistics in Medicine* 43, 13 (2024), 2672–2694.
- [2] Aad W Van der Vaart. 2000. *Asymptotic statistics*. Vol. 3. Cambridge university press.

Method	Goodreads						MovieLens-10M						Steam					
	AP (%)			AUC (%)			AP (%)			AUC (%)			AP (%)			AUC (%)		
	>0	>1	>2	>0	>1	>2	>0	>1	>2	>0	>1	>2	>0	>1	>2	>0	>1	>2
ATE	88.63	84.77	67.06	84.81	88.75	86.43	27.94	24.07	19.25	70.58	82.62	87.42	38.35	32.60	24.35	72.94	78.18	82.56
STE	89.14	86.14	68.60	85.09	89.43	87.24	28.70	26.69	21.34	69.69	83.27	87.96	50.53	47.76	39.90	76.51	81.74	86.08
Pearson	52.07	47.95	37.33	44.97	60.38	66.56	20.89	21.69	13.86	61.55	76.28	75.04	31.74	27.34	19.50	65.01	71.60	73.21
DoC	39.72	30.50	20.38	22.52	32.59	39.73	14.41	9.57	6.22	53.23	58.21	53.99	18.44	11.68	6.39	49.22	49.30	46.47
ABLT	86.39	84.34	63.54	82.67	87.90	84.94	21.44	20.62	13.17	65.19	75.17	76.80	37.08	29.39	22.64	74.12	78.39	82.34
ABLT/STD	88.20	85.26	68.15	84.17	88.71	86.62	24.99	20.84	16.40	65.73	79.86	83.41	47.42	42.85	34.20	77.03	82.05	85.75
Cosine	52.37	45.81	32.67	45.64	56.16	58.75	27.02	20.43	9.96	72.08	72.83	66.04	35.94	26.93	19.21	72.94	73.30	73.38
Jaccard	52.38	47.33	36.66	45.87	58.73	64.37	13.26	9.46	7.03	52.13	63.12	61.86	31.70	27.13	19.21	65.13	70.83	71.85
ATE (stabilized)	85.25	82.85	62.58	81.98	87.18	84.53	21.43	20.57	13.13	65.20	75.21	76.83	36.59	29.04	22.42	74.16	78.87	83.04
STE (stabilized)	88.10	84.83	67.91	84.06	88.39	86.41	24.97	20.77	16.37	65.74	79.86	83.39	45.69	41.04	33.52	76.82	82.12	86.04

Table 3: Main results on real-world organic CF logs. Each method induces a ranking over directed item pairs ($i \rightarrow j$) and is evaluated using LLM-based ordinal judgments as an external causal-plausibility signal (Sec. ??). For each threshold $t \in \{0, 1, 2\}$, positives satisfy $\text{causal_effect} > t$. We report Average Precision (AP) and ROC-AUC; the best result in each dataset-threshold block is bold. STE obtains the highest AP in all nine dataset-threshold settings.

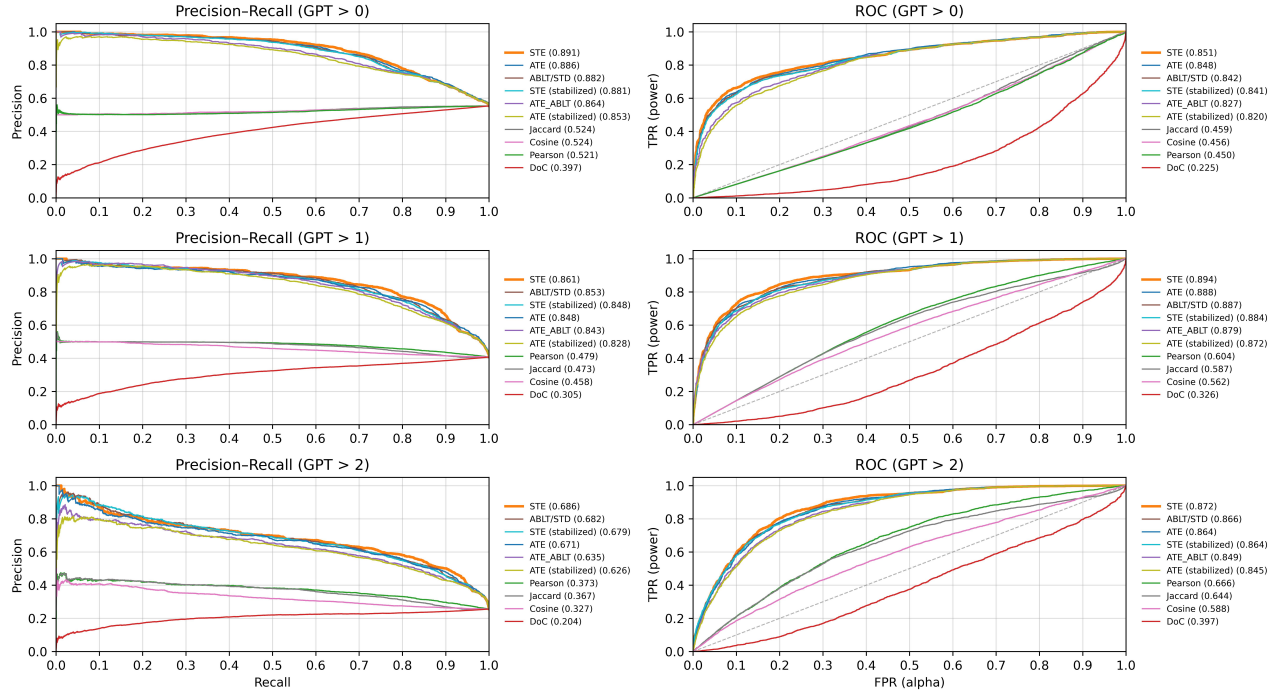


Figure 2: Goodreads PR/ROC. Precision-Recall and ROC curves for ranking directed item pairs. Positives are defined as $\text{causal_effect} > t$ for $t \in \{0, 1, 2\}$ using LLM-assigned causality labels. Legends report AP and ROC-AUC. As in Steam and MovieLens-10M, STE yields the strongest Precision-Recall trade-off.

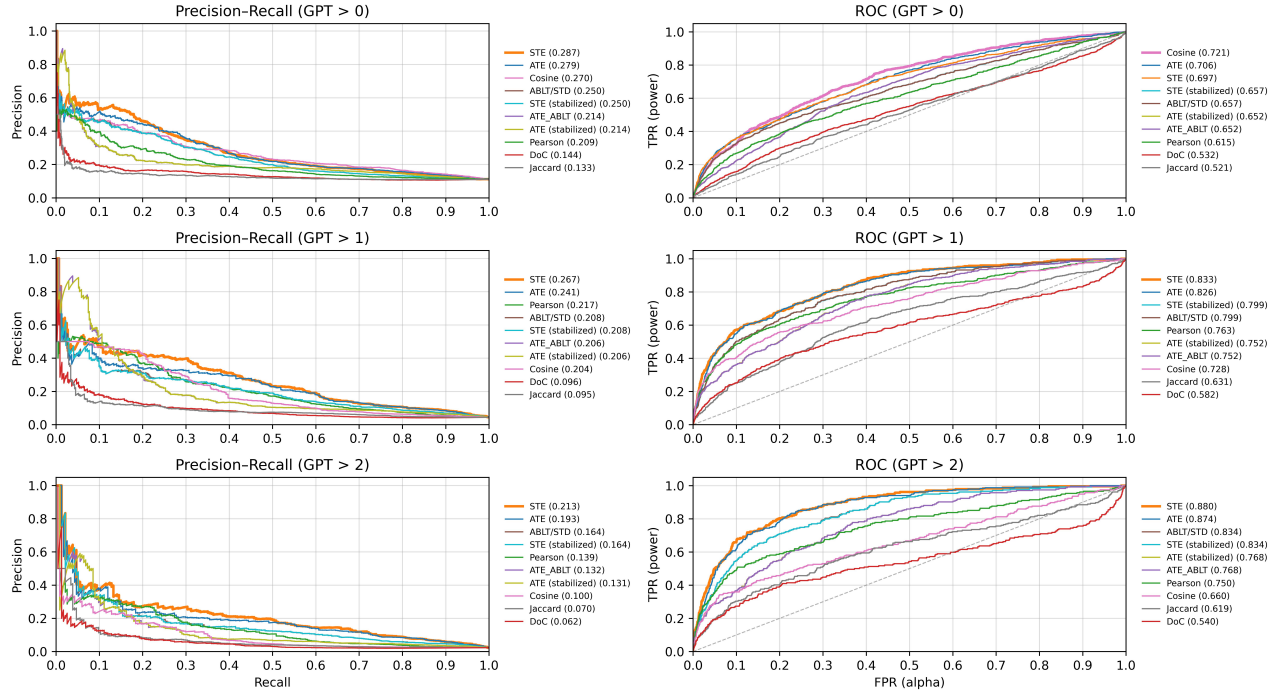


Figure 3: MovieLens-10M PR/ROC. Precision-Recall and ROC curves for ranking directed item pairs. Positives are defined as $\text{causal_effect} > t$ for $t \in \{0, 1, 2\}$ using LLM-assigned causality labels. Legends report AP and ROC-AUC. As in the main paper, STE provides the strongest Precision-Recall performance, especially at stricter thresholds.

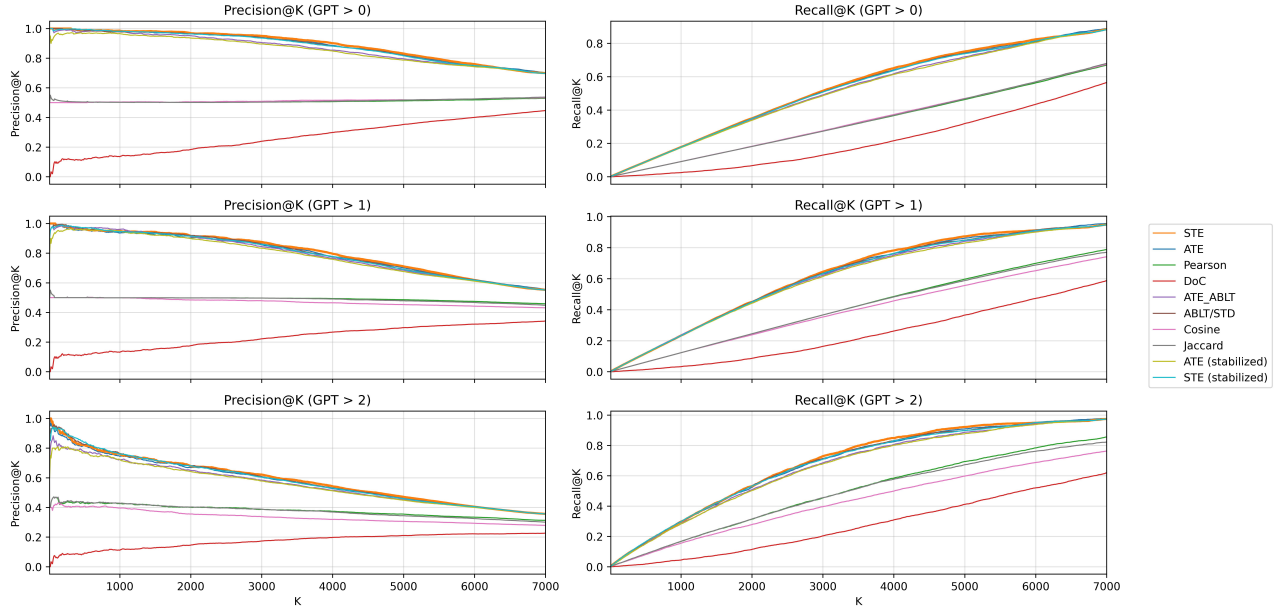


Figure 4: Goodreads Precision@K/Recall@K. Top-K evaluation using LLM-assigned causality labels, with positives defined as $\text{causal_effect} > t$ for $t \in \{0, 1, 2\}$. STE improves the quality of the highest-ranked candidate links, especially under the stricter thresholds.

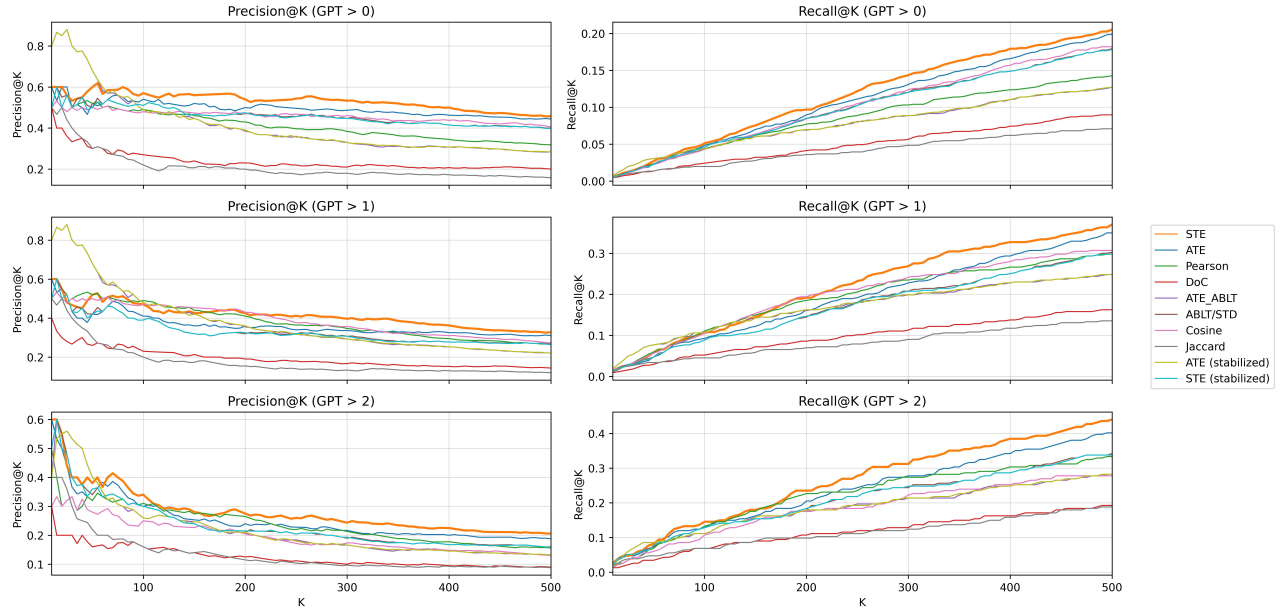


Figure 5: MovieLens-10M Precision@K/Recall@K. Top- K evaluation using LLM-assigned causality labels, with positives defined as $\text{causal_effect} > t$ for $t \in \{0, 1, 2\}$. Consistent with the AP results, STE gives the strongest top-ranked discovery quality.

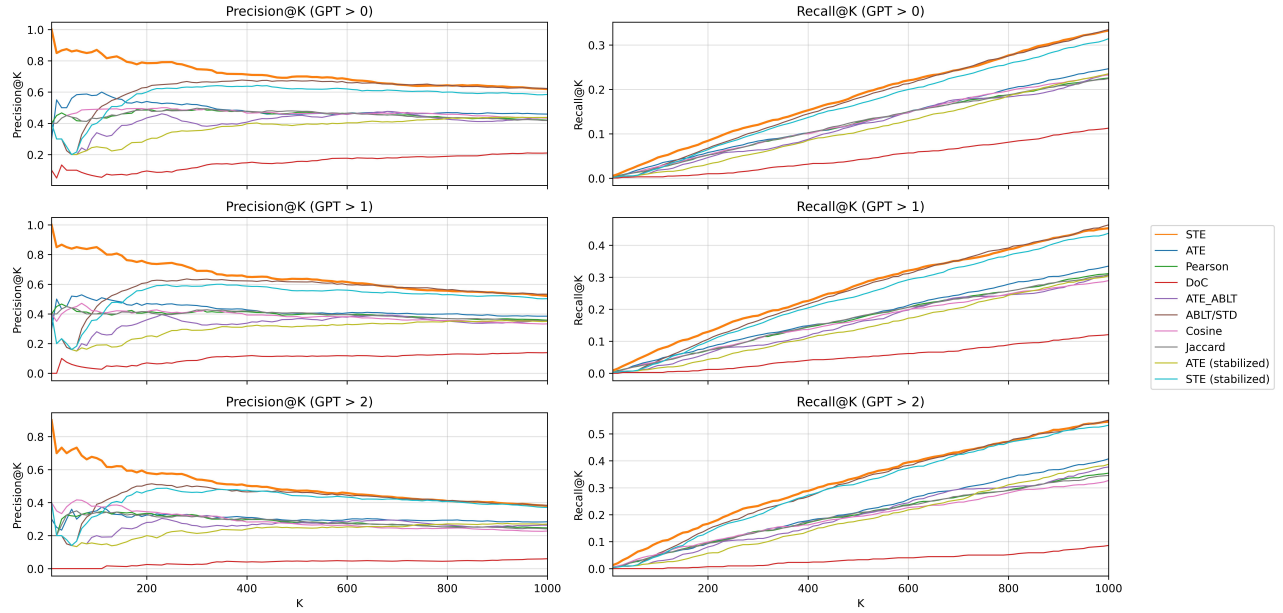


Figure 6: Steam Precision@K/Recall@K. Top- K evaluation using LLM-assigned causality labels, with positives defined as $\text{causal_effect} > t$ for $t \in \{0, 1, 2\}$. STE yields the most actionable top- K candidate sets in the higher-confidence regimes.